Hypothesis Testing for Correlation

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Hypothesis testing for correlation

You should be familiar with using a **hypothesis test** to determine bias within probability problems. It is also possible to use a hypothesis test to determine whether a given **product moment correlation coefficient** calculated from a **sample** could be representative of the same relationship existing within the **whole population.** For full information on hypothesis testing, see the revision notes from section 5.1.1 Hypothesis Testing

Why use a hypothesis test for correlation?

- In most cases it is too difficult to get the value of the PMCC for a whole population
  - This would involve having data on each individual within the whole population
  - It is very rare that a statistician would have the time or resources to collect all of that data
- The PMCC for the whole population can instead be estimated using information from a sample taken from the population
  - The PMCC for a whole population is denoted ρ (pronounced rho)

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

of r to determine whether the population can be said to have positive, negative or zero correlation

How do I use the table of critical values for correlation coefficients?

- The table of critical values is found near the end of the Mathematical Formulae and Statistical Tables booklet you will have in your exam
- It displays the minimum values that need to be reached by a value of r in order for the test to be significant
- Find the significant value by looking at the value that is in both the row of the number in the sample, n, and in the column for the significance level of the test
  - Values are given for sample sizes of $4 \leq n \leq 30$ and then 40, 50, 60, 70, 80, 90 and 100
  - Values are given for significance levels of 10% (0.1), 5% (0.05), 2.5% (0.025), 1% (0.01) and 0.5% (0.005)

- The table gives values for one - tailed tests only
  - To find the significant values for two - tailed tests halve the significance level first and then carry out the test as if doing a one-tailed test

- The table gives values for positive values of r only
  - To find the significant values if you are testing for negative correlation look for the corresponding positive value of r and change the sign to a negative
  - A two-tailed test will have two critical values (both the same value but one positive and one negative)

How do I lay out a hypothesis test for correlation?

- Most of the time the hypothesis test will be carried out by using a **critical value**
- You won't be expected to calculate p-values but you might be given a p-value
- **Step 1.** Write the null and alternative hypotheses clearly
  - The hypothesis test could either be a **one-tailed test** or a **two-tailed test**
  - The null hypothesis will always be $H_0: \rho = 0$
  - The alternative hypothesis will depend on if it is a one-tailed or two-tailed test
    - A one-tailed test would test to see if the **population PMCC, ρ,** is **either positive or negative**
      - The **alternative hypothesis, H_1 will be** $H_1: \rho > 0$ **or** $H_1: \rho < 0$

    - A two-tailed test would test to see if the **population PMCC, ρ** , is **not equal to zero** (meaning there is some form of linear correlation)
      - The **alternative hypothesis, H_1 will be** $H_1: \rho \neq 0$

- **Step 2.** Find the critical value for the test
  - The **critical value** will be determined by both the **significance level** of the test and the **sample size, n**
    - For a positive value of r :

        - The greater the significance level the lower r needs to be to determine that ρ is also positive
    - The **critical value** for can be found by using the **table of critical values** given in the Mathematical Formulae and Statistical Tables booklet
        - Make sure you know how to find this and are familiar with using it before your exam
- **Step 3.** Compare the value of r calculated from the sample with the critical value
    - If r is in the critical region the test is **significant** and the null hypothesis should be rejected
        - It will be in the critical region if $|r| > |\text{critical value}|$
    - If r is not in the critical region the null hypothesis should be accepted and the alternative hypothesis should be rejected
- **Step 4.** Write a conclusion in context
    - Use the wording in the question to help you write your conclusion
    - If rejecting the null hypothesis your conclusion should state that there is **sufficient evidence to suggest** the alternative hypothesis is true at this level of significance
    - If accepting the null hypothesis your conclusion should state that there is **not enough evidence** to suggest the alternative hypothesis is true at this level of significance
- Occasionally the hypothesis test will be carried out by comparing a **p – value** with the significance level instead
    - You will not be expected to calculate the p - value, it will be given in the question
    - Steps 1 and 4 will be the same, however you should compare the p - value with the significance level in step 3 (there is no step 2)
        - If the p - value is less than the significance level the test is **significant** and the null hypothesis should be rejected
        - If the p - value is greater than the significance level the null hypothesis should be accepted and the alternative hypothesis should be rejected

Worked Example

A student believes that there is a positive correlation between the number of hours spent studying for a test and the percentage scored on it.

The student takes a random sample of 10 of his friends and records the amount of revision they did and percentage they score in the test.

The student calculates the product moment correlation coefficient for these data as $r = 0.668$.

Answer:

Examiner Tips and Tricks

- Make sure you read the question carefully to determine whether the test you are carrying out is for a one-tailed or a two-tailed test and use the level of significance accordingly. A test is only one-tailed if you are told to test for positive or negative correlation. If the question says test for correlation, then it is a two-tailed test, even if you think the correlation would be positive.

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